Learning to Distrust Misleading Text: Preference Optimization for Reducing Text Bias in Vision-Language Models

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Abstract

Vision-language models (VLMs) often treat textual context as a shortcut, even when that text conflicts with visual evidence. This "blind faith in text" is especially problematic in document and visual question answering, where auxiliary text may be incomplete, stale, adversarial, or automatically retrieved. We study whether preference optimization can reduce this failure mode without sacrificing normal visual question answering performance. Starting from corrupted-text evaluations, we mine model-specific text-following errors: cases where the model is wrong and its wrong answer is copied from the corrupted auxiliary text. We then train VLMs with Direct Preference Optimization (DPO), preferring the image-grounded answer over the model's own text-following error. On DocVQA with Qwen3-VL-8B, text-following span DPO improves corrupted accuracy from 0.715 to 0.915 and reduces the fraction of incorrect answers copied from corrupted text from 89.5% to 47.1%. A multi-condition error-mined extension further improves accuracy to 0.920 and reduces this text-following rate to 43.8%. By contrast, GRPO-style sampled-answer optimization and auxiliary SFT preservation do not improve over DPO. Experiments on VQAv2 and transfer to Qwen2-VL and LLaVA-Next show that the method works best when the base VLM already has sufficient visual and OCR competence. These results suggest that high-precision preference pairs are an effective and practical mechanism for correcting text bias in VLMs, while also revealing clear boundary conditions for the approach.

1. Introduction

Vision-language models (VLMs) are increasingly used in settings where images are accompanied by textual context: document images with OCR, screenshots with metadata, retrieved descriptions, captions, or user-provided hints. Such text can be helpful when it is correct, but it can also become a shortcut. When textual context conflicts with the image, a VLM may answer from the text rather than from visual evidence. Deng et al. [2] describe this phenomenon as "blind faith in text", showing that several VLMs disproportionately trust textual information under conflict.

This failure is not merely a benchmark artifact. In practical systems, auxiliary text is often produced by retrieval, OCR, captioning, or prior model predictions. All of these sources can be noisy. If a VLM over-trusts such text, the model can produce confident but visually ungrounded answers. For document VQA, this is particularly important: the model must decide whether the image or the accompanying text should be trusted when they disagree.

We ask a targeted question: *Can we reduce text-following failures by directly training a VLM to prefer image-grounded answers over its own corrupted-text errors?* Our key observation is that the corrupted-text setting naturally produces preference pairs. If a model answers incorrectly and its prediction appears in the corrupted auxiliary text, then the model has likely followed the misleading text. The ground-truth answer can be used as the preferred response, and the model's text-following prediction can be used as the dispreferred response. This gives a high-precision preference signal without manual labeling.

We instantiate this idea with Direct Preference Optimization (DPO) [7]. Unlike online reinforcement learning methods such as PPO or GRPO [8], DPO directly optimizes pairwise preferences. This is well aligned with our setting: the contrastive pair is already available as {image-grounded answer, text-following wrong answer}. We compare DPO against GRPO-style sampled-answer optimization, preservation SFT, model transfer, and dataset transfer.

Our study makes three empirical findings. First, on DocVQA with Qwen3-VL-8B, DPO sharply improves corrupted-text robustness while reducing the rate of text-copying among remaining errors. Second, broadening DPO from corrupted-only text-following errors to multi-condition error-mined pairs yields the strongest result, suggesting that condition-specific error mining can improve stability. Third, the method has an important boundary condition: it works best when the base VLM already has enough visual/OCR competence. Qwen-family models benefit substantially, while LLaVA-Next models under our prompt and template show weak DocVQA base performance and limited benefit.

We summarize our contributions as follows:

- We define an automatic text-following error mining procedure for corrupted-text VQA, requiring both incorrectness and overlap between the wrong prediction and corrupted auxiliary text.
- We show that DPO on these mined pairs substantially reduces text bias on DocVQA and transfers directionally to VQAv2 and Qwen2-VL.
- We compare DPO, GRPO, SFT preservation, multi-condition error mining, and model transfer, clarifying when preference optimization helps and when it does not.

2. Related Work

Text bias in VLMs. VLMs are trained to fuse visual and textual inputs, but fusion can become asymmetric when textual context is easier to exploit than visual evidence. The most directly related work is Words or Vision [2], which introduces corrupted textual variations and shows that VLMs may trust text over vision when the two conflict. Our work follows this diagnostic setting but focuses on mitigation through preference optimization. Rather than only measuring the degradation under corrupted text, we use the model's own corrupted-text failures as training signal.

Document and visual question answering. DocVQA [6] evaluates question answering over document images, requiring OCR, layout understanding, and reasoning over visual text. VQAv2 [3] evaluates open-ended visual question answering with reduced language priors compared to earlier VQA datasets. We use DocVQA as the main testbed because conflict between visual document evidence and auxiliary text is central to the task. VQAv2 is used as a transfer check to test whether the mitigation is specific to document images.

Preference optimization. Preference optimization has become a common alternative to supervised fine-tuning for aligning generative models. DPO [7] directly optimizes a policy from chosen/rejected pairs without training an explicit reward model or running online RL. This makes it attractive for our setting because corrupted-text evaluation yields natural preference pairs. GRPO [8] is an online reinforcement learning method that estimates a group-relative baseline from multiple sampled responses. We compare DPO with GRPO-style optimization and find that free-form sampled-answer rewards are less effective for this failure mode.

Modern VLMs. Qwen2-VL [9] and Qwen2.5-VL [1] emphasize dynamic-resolution visual processing and strong document/OCR capability. LLaVA-style models [4, 5] provide a widely used visual instruction tuning framework. Our transfer experiments suggest that these model families differ substantially in the base competence needed for text-bias correction.

3. Method

3.1. Problem Setup

Each example consists of an image x, a question q, auxiliary text t, and a ground-truth answer y. We consider three main textual conditions. In the *match* condition, the auxiliary text is consistent with the image. In the *irrelevant* condition, the text is unrelated. In the *corrupted* condition, the text contains a plausible but wrong answer. The desired behavior is not to ignore text unconditionally. Instead, the model should use text when it is reliable and fall back to visual evidence when text conflicts with the image.

3.2. Mining Text-Following Errors

We first evaluate a base VLM on corrupted examples. Let $\hat{y}$ be the model prediction. We define a *text-following error* when all of the following hold:

1. the model prediction is incorrect;
2. the prediction is non-empty and does not match the ground truth;
3. the prediction appears in the corrupted auxiliary text.

For the third condition, we use normalized string matching. For multi-token answers, we also accept high token overlap with the auxiliary text. This filtering is important: an arbitrary wrong answer is not necessarily a text-bias error, but an incorrect prediction copied from corrupted text is a much stronger signal.

For each mined error, we construct a DPO pair:

$$y^+ = y, \qquad y^- = \hat{y}, \tag{1}$$

where y^+ is the image-grounded answer and y^- is the model's text-following wrong answer. We call this dataset *text-following span DPO*.

3.3. DPO Objective

Given a policy π_θ and a reference policy π_{ref}, DPO optimizes the preference that y^+ should be more likely than y^- under the prompt (x, q, t):

$$\begin{aligned}\mathcal{L}_{\text{DPO}} = -\log\sigma\Big(\beta\big[\log\pi_\theta(y^+|x,q,t) - \log\pi_{\text{ref}}(y^+|x,q,t) \\ - \log\pi_\theta(y^-|x,q,t) + \log\pi_{\text{ref}}(y^-|x,q,t)\big]\Big).\end{aligned} \tag{2}$$

In our setting, this objective directly encodes the desired correction: the image-grounded answer should be preferred over the corrupted-text answer.

3.4. Multi-Condition Error-Mined DPO

Text-following span DPO uses only corrupted examples. To reduce drift and improve stability, we also test a multi-condition extension. We mine errors from corrupted, match, irrelevant, and no-text evaluations. For corrupted examples, we keep the strict text-following filter: the wrong prediction must appear in corrupted auxiliary text. For match, irrelevant, and no-text examples, we use a broader condition-specific error filter: the prediction must be incorrect, non-empty, and distinct from the ground truth.

We assign lower weights to non-corrupted errors:

$$\begin{aligned} w_{\text{corrupted}} &= 1.0, \quad w_{\text{match}} = 0.25, \\ w_{\text{irrelevant}} &= 0.5, \quad w_{\text{no-text}} = 0.5. \end{aligned} \tag{3}$$

This preserves the central text-bias signal while adding weak regularization from other conditions.

3.5. GRPO-Style Baselines

We also test GRPO-style optimization. For each prompt, the model samples multiple candidate answers. A reward function gives positive reward for matching the chosen answer and penalties for matching the rejected answer, misleading spans mined from corrupted text, or any answer copied from the auxiliary text. The reward also mildly encourages short answer-like completions. This tests whether online sampled-answer optimization can discover the same correction signal without explicit pairwise DPO.

4. Experiments

4.1. Metrics

We report three complementary metrics. *Corrupted accuracy* measures task performance when auxiliary text is misleading. *Incorrect aux-hit* measures the fraction of incorrect predictions that appear in the corrupted auxiliary text. Lower incorrect aux-hit indicates less text-following among remaining errors, but it is not sufficient by itself: a model could reduce aux-hit by hallucinating unrelated answers. Therefore, we interpret it together with corrupted accuracy and preservation on match/irrelevant conditions. We also report corrected/regressed counts relative to the base model when available.

4.2. DocVQA Main Results

DocVQA is our main testbed. We use Qwen3-VL-8B-Instruct as the primary model and evaluate on a held-out corrupted split. Table 1 shows that text-following span DPO substantially improves both accuracy and text-bias behavior. Increasing LoRA rank from 8 to 16 improves the result further, while the multi-condition error-mined variant gives the best overall performance.

Table 1. DocVQA corrupted held-out results with Qwen3-VL-8B.

Method	Acc.	Incorrect aux-hit	Corr./Reg.
Base	0.715	89.5%	–
TF-span DPO, r8	0.900	55.0%	37 / 0
TF-span DPO, r16	0.915	47.1%	40 / 0
Multi-cond. error DPO	**0.920**	**43.8%**	41 / 0

Table 2. GRPO-style optimization does not improve over DPO on DocVQA.

Variant	Init	Acc.	Incorrect aux-hit
Base	–	0.715	89.5%
GRPO-only, 100 steps	base	0.710	89.7%
GRPO-only, 300 steps	base	0.705	89.8%
GRPO-only, dense reward	base	0.705	89.8%
TF-span DPO	base	**0.900**	**55.0%**
DPO → GRPO	DPO	**0.900**	60.0%

The multi-condition result suggests that the highest-performing variant is not corrupted-only. However, the corrupted filter remains essential: corrupted rows still require the prediction to be copied from corrupted auxiliary text. The additional match, irrelevant, and no-text errors act as weak condition-specific regularizers rather than replacing the text-following signal.

4.3. DPO versus GRPO

We compare DPO with GRPO-only and DPO-to-GRPO variants on the same disjoint DocVQA split. GRPO samples multiple answers per prompt and optimizes a reward that encourages correctness and penalizes copying corrupted text. Despite this denser reward, GRPO does not move the held-out greedy behavior meaningfully. Table 2 shows that DPO is the effective intervention in our setting.

DPO-to-GRPO changes only a few held-out predictions relative to DPO and does not improve accuracy. This supports our hypothesis: when high-precision chosen/rejected pairs are available, direct pairwise optimization provides a clearer learning signal than free-form sampled-answer rewards.

4.4. VQAv2 Transfer Check

We repeat the disjoint split protocol on VQAv2. The effect is positive but weaker than on DocVQA. As shown in Table 3, span DPO improves corrupted accuracy by 10 points and reduces incorrect aux-hit, but match and irrelevant preservation drop slightly. Adding auxiliary SFT preservation does not recover this drop and slightly worsens corrupted performance.

Table 3. VQAv2 transfer results with Qwen3-VL-8B.

Variant	Corr. Acc.	Aux-hit	Match	Irr.
Base	0.545	65.9%	0.92	0.77
Span DPO	**0.645**	**40.8%**	0.90	0.75
Span DPO + SFT	0.635	42.5%	0.90	0.75

Table 4. Model transfer on DocVQA. DPO helps Qwen2-VL but not LLaVA-Next under our setting.

Model	Setting	Acc.	Aux-hit	Corr./Reg.
Qwen2-VL-7B	base	0.570	93.0%	–
Qwen2-VL-7B	span DPO	**0.775**	86.7%	41 / 0
LLaVA-N-7B	base	0.105	90.5%	–
LLaVA-N-7B	span DPO	0.100	91.7%	1 / 2
LLaVA-N-13B	base	0.115	92.7%	–
LLaVA-N-13B	span DPO	0.130	95.4%	3 / 0

4.5. Model Transfer

We test whether the same mitigation extends beyond Qwen3-VL. Table 4 summarizes Qwen2-VL-7B and LLaVA-Next results on DocVQA. Qwen2-VL benefits substantially, improving from 0.570 to 0.775 corrupted accuracy. LLaVA-Next, however, has very low base DocVQA accuracy under our prompt/template, and DPO provides little benefit.

These results show that the method is not simply a universal post-hoc fix. It requires a base model that can already solve the task visually; DPO then teaches the model to resist misleading text.

5. Analysis and Discussion

5.1. Why DPO Fits This Problem

The corrupted-text setting gives unusually clean preference pairs. For many failures, the model's wrong answer is directly copied from the corrupted auxiliary text. Thus the comparison is explicit: the image-grounded answer should be preferred over the text-following wrong answer. DPO optimizes exactly this comparison. In contrast, GRPO must first sample useful candidate answers and then rely on a hand-designed reward. If the sampled candidates rarely contain the correct answer, or if most candidates are similar text-following errors, the relative reward signal is weak. This explains why GRPO logs can show diagnostic reward events while greedy held-out predictions barely change.

5.2. Interpreting Incorrect Aux-Hit

Incorrect aux-hit is useful but should not be optimized alone. A lower value means fewer remaining errors are copied from corrupted text, but it does not guarantee better visual grounding. For example, a model could avoid the corrupted text yet hallucinate an unrelated answer. Therefore, successful text-bias correction should satisfy three conditions:

1. corrupted accuracy should increase;
2. incorrect aux-hit should decrease;
3. match and irrelevant performance should be preserved.

Our main DocVQA results satisfy these conditions. Some negative ablations do not: they reduce neither error copying nor accuracy, or they preserve match performance while failing on corrupted examples.

5.3. Why Multi-Condition Error Mining Helps

Text-following span DPO is the core filtering strategy, but the best result comes from multi-condition error-mined DPO. This does not mean that unfiltered errors are sufficient. Rather, corrupted errors remain strictly filtered for text-following behavior, while other conditions contribute lower-weight stability signals. This helps prevent the model from over-specializing to the corrupted prompt format. The gain is modest because non-corrupted failures are sparse in DocVQA, but it consistently improves the main held-out metric.

5.4. Boundary Conditions

Our transfer experiments reveal a clear boundary condition. The method works when the base VLM already has sufficient visual and OCR competence. Qwen-family models satisfy this condition on DocVQA and improve substantially. LLaVA-Next models, under our prompt/template, have very low base DocVQA accuracy. In this regime, DPO cannot create document-reading ability from scratch; it can only alter preferences among answers the model can already represent.

TextVQA experiments lead to a related conclusion. DocVQA-trained DPO transfers weakly to TextVQA and does not reliably fix LLaVA-specific TextVQA conflict failures. Even after mining larger LLaVA-specific TextVQA DPO data and adding strict+preserve pairs, corrupted performance does not improve. This suggests that TextVQA conflict involves different OCR and answer-format dynamics than DocVQA. Therefore, we treat TextVQA and LLaVA results as boundary analyses rather than main success claims.

5.5. Limitations

Our study has several limitations. First, the text-following filter is an operational definition, not a causal proof. If a wrong answer appears in corrupted text, the model likely followed the text, but accidental overlap is possible. Second, most experiments use a single held-out split and a limited number of seeds due to compute constraints. Third, our GRPO implementation uses free-form sampled answers;

a candidate-constrained GRPO-style objective may be a stronger comparison. Finally, the method assumes ground-truth answers are available for preference construction. This is suitable for benchmark adaptation and supervised mitigation, but not directly for fully unlabeled deployment.

6. Conclusion

We studied preference optimization as a mitigation for VLM text bias under corrupted auxiliary text. By mining text-following errors and training with DPO, we convert a diagnostic failure mode into a direct preference-learning signal. On DocVQA, this substantially improves corrupted-text accuracy and reduces the rate at which remaining errors copy misleading text. The strongest variant, multi-condition error-mined DPO, combines strict corrupted-error filtering with lower-weight errors from match, irrelevant, and no-text conditions. GRPO-style sampled-answer optimization and preservation SFT do not improve over DPO in our experiments. The results support a precise conclusion: DPO is effective when the base VLM already has visual/OCR competence and the preference pairs accurately capture text-following failures. When this condition is not met, as in our LLaVA-Next DocVQA setting, preference optimization alone is insufficient.

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